TKR COLLEGE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS

Department of Computer Science and Engineering

Project Domain:

Artificial Intelligence and Machine Learning (AI & ML)

Project Title:

Pneumonia Detection Using Deep Learning

ABSTRACT

Pneumonia still is one of the biggest killers globally and especially for the vulnerable population groups. Traditional methods of diagnosis still take a lot of time, hence delaying important treatments while in the process; they can also be inaccurate. There is an enormous capability for advanced artificial intelligence surveillance that can bring about dramatically improved diagnosis of pneumonia through assessing X-ray scans within the shortest time possible. However, previous literature shows that this remains a difficult task, which is why such an approach is needed. This includes training with different data, as well as the differentiation of viral and bacterial pneumonia, normalization, and augmentation methods. Thus, through the creation of an open-source web tool that implements these solutions, not only the medical industry is progressed, but the individual who is in need of this tool can have a chance to save a life. This project's success would mean a considerable decision regarding the future of AI and the strong possible lifesaving potential it has when applied to medical practice in general and pneumonia diagnosis in particular. The seriousness and purpose of this endeavor are that, through applying such best practices, positive changes to health care systems at an international level can be achieved.

Keywords:

Pneumonia Detection, Chest X-Ray Images, Deep Learning, Artificial Intelligence, Machine Learning, Convolutional Neural Network, U-Net, Lung Segmentation, Medical Image Processing, Heatmap Visualization, Clinical Decision Support.

Project Definition:

The project aims to develop an intelligent pneumonia detection system that analyzes chest X-ray images using deep learning. The system applies image preprocessing, normalization, augmentation, and lung segmentation to focus on the important lung regions. A deep learning model is then used to classify the X-ray image into normal, bacterial pneumonia, or viral pneumonia categories. The system also generates heatmap-based visual explanations to highlight affected lung areas, making the prediction more transparent and useful for medical decision support.

Objectives:

1. Develop a deep learning-based model for pneumonia detection from chest X-ray images.
2. Apply preprocessing, normalization, and augmentation techniques to improve X-ray image quality.
3. Use U-Net for lung segmentation to remove unnecessary background regions.
4. Classify chest X-ray images into normal, bacterial pneumonia, and viral pneumonia categories.
5. Generate heatmap visualizations to explain the affected lung regions.
6. Develop a simple web-based tool for uploading X-ray images and displaying prediction results.
7. Evaluate system performance using Accuracy, Precision, Recall, and F1-Score.

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